

	Marks	Credits
L T P	Exam: 100	4
4 - -	Sessional: 50	
	Total: 150	4

Duration of Exam: 3 hrs.

Course Outcomes:

At the end of the course the student will:

CO1. Be able to understand the knowledge-based systems representation.

CO2. Be able to understand automatic reasoning.

CO3. Be able to understand inductive and deductive learning.

CO4. Be able to implement a small knowledge- based system

NOTE:

Examiner will set nine questions in total. Question One will be compulsory and will comprises of all sections and remaining eight questions to be set by taking two questions from each unit. The students have to attempt five questions in total, first being compulsory and selecting one from each Unit.

UNIT-I

Introduction to Logic, Propositional Logic concepts, SemanticTebbleaux and Resolution in Propositional Logic, FOPL, SemanticTebbleaux in Predicate Logic, and Resolution in Predicate Logic, Logic Programming in Prolog.

UNIT-II

Knowledge Representation, Semantic Nets, Partitioned Nets, Parallel Implementation of Semantic Nets. Frames, Common Sense Reasoning and Thematic Role Frames, Architecture of Knowledge Based System, Rule Based Systems,Framebasedsystems. Forward and Backward Chaining,

UNIT-III

Search Techniques. Uninformed Search: DFS, BFS, Iterative Deepening, Heuristic Search: A*, Hill Climbing etc.

UNIT-IV

Uncertainty Management in Expert Systems, Fuzzy Logic, ProbabilisticMethods, Bayesian Theory, Dempster Shafer Theory, Bayes Network, Introduction to Agents and their Application in Intelligent Systems.

References:

1. Artificial Intelligence-Nils J Nilson
2. Artificial Intelligence-Elain Rich and Kevin Knight
3. Artificial Intelligence: A modern approach-Staurt Russel and Peter Norvig
4. Artificial Intelligence-Patrick Henry Winston
5. The Essence of Logic- John Kelly